

Hermes Architecture

Hermes Agent Architecture

Public-safe architecture snapshot generated at 2026-06-27T06:15:24-04:00.

Source of truth: local Hermes configuration and runtime status on the operator Linux host.

Sensitive values, private chat IDs, OAuth headers, personal finance/client workflow names, and tokens are intentionally omitted or grouped.

Executive Summary

Hermes Agent runs as a **Linux-primary, multi-platform AI operations hub**. The gateway is a systemd service that connects Telegram, Mattermost, WhatsApp, API Server, Home Assistant, MCP servers, cron jobs, skills, memory/context stores, and remote Windows operations.

The default model remains **openai-codex / gpt-5.5**. Local/experimental providers such as **LM Studio** and **Free Kimi** are configured as optional providers only; they are not the default route for Telegram or production workflows.

High-Level Architecture

[!Hermes Architecture](#)

Runtime Topology

Layer	Components	Notes
User channels	Telegram, Mattermost, WhatsApp, API Server	Gateway adapters route inbound messages into the same Hermes agent core.
Agent core	Hermes Agent gateway + CLI + tool loop	Runs on Linux under <code>hermes-gateway.service</code> .
Models	OpenAI Codex primary, Copilot fallback, optional LM Studio, optional Free Kimi proxy	Provider selection is explicit; experimental endpoints are not default.
Tools	Terminal, files, browser, web, vision, TTS, cron, delegation, Home Assistant, GBrain, NotebookLM, CodeGraph	Enabled toolsets differ by platform/session but the gateway exposes the main operational surface.
Knowledge	GBrain, NotebookLM, session search, imported Claude context, skills	Long-term memory and research context are split across structured stores and markdown skills.
Automation	Hermes cron scheduler, no-agent scripts, hooks, plugins	Backups, monitoring, GitHub publication, finance snapshots, smart-home logging, news/media digests, and watchdogs.
Operational surfaces	Tasks, skills, hooks, plugins, MCP servers, toolsets	Documented as public-safe counts/categories plus selected non-sensitive examples.
Remote systems	Windows workstation over SSH, Home Assistant, Google Drive/GitHub, local LM Studio	Linux remains the control plane; Windows is operated remotely when needed.

Operational Surface Inventory

Surface	Detected public-safe state	Notes
Scheduled tasks / cron	39 jobs; 27 no-agent script jobs; 0 agent-backed jobs	Exact private task names are grouped by category.
Skills	227 detected skill files across 22 categories	Private/client-sensitive skill names are omitted from examples.
Hooks / webhooks	shell allowlist present: False; allowlist entries: 0; plugin hook manifests: 29	Hook command bodies are not published.
Plugins	75 visible plugin rows captured; enabled estimate 4	Descriptions omitted to avoid leaking credential/env surfaces.
MCP servers	8 configured MCP servers	GBrain, NotebookLM, CodeGraph are the active core MCP surfaces.

Scheduled tasks / cron categories

Category	Active jobs	Public-safe purpose
Backup & sync	5	Protect configuration, repositories, databases, and knowledge stores.
GitHub & publishing	6	Maintain GitHub/publication surfaces and repo health digests.
Home automation	2	Log smart-home/home-environment telemetry.
Knowledge & memory	6	Keep GBrain/memory/context stores healthy and up to date.
Media/news monitoring	3	News, RSS, YouTube, and briefing pipelines.
Other scheduled automation	5	Other local automation jobs.
Private finance automation	5	Private finance workflow snapshots; details omitted from public docs.
Reliability watchdogs	7	Auto-healing, environment guards, timeout/watchdog checks.

Skills surface

Hermes currently has a broad skill surface. The public inventory lists category counts and non-sensitive examples only.

Skill category	Count
.archive	11
apple	5
autonomous-ai-agents	7
creative	34
data-science	2
devops	8
ecc-imports	4
email	2
gaming	2
github	11
mcp	1
media	7
mlops	19
note-taking	4
personal	6
productivity	23
red-teaming	1
research	15
smart-home	5
social-media	2
software-development	38
uncategorized	20

Public-safe skill examples:

Skill	Category	Public-safe description
dspy	mlops	DSPy: declarative LM programs, auto-optimize prompts, RAG.
fine-tuning-with-trl	mlops	TRL: SFT, DPO, PPO, GRPO, reward modeling for LLM RLHF.
axolotl	mlops	Axolotl: YAML LLM fine-tuning (LoRA, DPO, GRPO).
unsloth	mlops	Unsloth: 2-5x faster LoRA/QLoRA fine-tuning, less VRAM.
huggingface-hub	mlops	HuggingFace hf CLI: search/download/upload models, datasets.
segment-anything-model	mlops	SAM: zero-shot image segmentation via points, boxes, masks.
weights-and-biases	mlops	W&B: log ML experiments, sweeps, model registry, dashboards.
evaluating-llms-harness	mlops	lm-eval-harness: benchmark LLMs (MMLU, GSM8K, etc.).
llama-cpp	mlops	llama.cpp local GGUF inference + HF Hub model discovery.
obliteratus	mlops	OBLITERATUS: obliterate LLM refusals (diff-in-means).
serving-llms-vllm	mlops	vLLM: high-throughput LLM serving, OpenAI API, quantization.
outlines	mlops	Outlines: structured JSON/regex/Pydantic LLM generation.
strategic-reading	uncategorized	Read a book, article, transcript, or case study through the lens of a specific strategic problem you
gif-search	media	Search/download GIFs from Tenor via curl + jq.
spotify	media	Spotify: play, search, queue, manage playlists and devices.
youtube-content	media	YouTube transcripts to summaries, threads, blogs.
article-enrichment	uncategorized	Transform raw article text dumps in the brain into structured pages with executive summary, verbatim quotes, key insights, why-it-matters, a
llm-wiki	research	Karpathy
pixelrag-mcp-operations	research	Operate and harden a local PixelRAG MCP wrapper for Hermes without exposing dangerous long-lived serve helpers through the gateway.
blogwatcher	research	Monitor blogs and RSS/Atom feeds via blogwatcher-cli tool.
[REDACTED]	research	Evaluate whether a third-party service or product is trustworthy, integrable, and compliant enough for automation, purchasing, or review for

Skill	Category	Public-safe description
[REDACTED]	research	Operate the academic research workflow from paper discovery through experiment-backed writing and submission.
polymarket	research	Query Polymarket: markets, prices, orderbooks, history.
skillpack-check	uncategorized	/
effective-liteparse	uncategorized	Use when a task involves a local PDF, DOCX, PPTX, XLSX, or image and you want the fastest local extraction path. Prefer <code>lit</code> before heavier

Hooks, webhooks, and plugin hook manifests

Hermes has multiple hook-related surfaces: shell-hook allowlists, webhook subscriptions, and imported Claude/Claude plugin hook manifests. The public repo records only surface/count information, not hook command bodies.

Hook manifest surface
<code>.hermes/oss-evals/2026-06-26/orca-sandbox/home/.gemini/config/hooks.json</code>
<code>.hermes/oss-evals/2026-06-26/orca-sandbox/home/.cursor/hooks.json</code>
<code>.hermes/oss-evals/2026-06-26/orca-sandbox/config/orca/codex-runtime-home/home/hooks.json</code>
<code>.hermes/hermes-agent/.codex/hooks.json</code>
<code>.claude/plugins/marketplaces/thedotmack/plugin/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/thedotmack/cursor-hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-plugins-official/plugins/learning-output-style/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-plugins-official/plugins/security-guidance/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-plugins-official/plugins/hookify/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-plugins-official/plugins/ralph-loop/hooks/hooks.json</code>
[REDACTED]
<code>.claude/plugins/marketplaces/claude-code-warp/plugins/warp/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/headroom-marketplace/plugins/headroom-agent-hooks/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-code-plugins/plugins/learning-output-style/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-code-plugins/plugins/security-guidance/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-code-plugins/plugins/hookify/hooks/hooks.json</code>
<code>.claude/plugins/marketplaces/claude-code-plugins/plugins/ralph-wiggum/hooks/hooks.json</code>
[REDACTED]
<code>.claude/plugins/marketplaces/ponytail/hooks/hooks.json</code>
<code>.claude/plugins/cache/thedotmack/claude-mem/13.2.0/hooks/hooks.json</code>

Plugin surface

Plugin	Status
browser-browser-use	not enabled
browser-browserbase	not enabled
browser-firecrawl	not enabled
chronos	not enabled
basic	not enabled
drain	not enabled
nous	not enabled
self-hosted	not enabled
disk-cleanup	not enabled
google_meet	not enabled
fal	not enabled
krea	not enabled
openai	not enabled
openai-codex	not enabled
openrouter	not enabled
xai	not enabled
alibaba-provider	not enabled
anthropic-provider	not enabled
arcee-provider	not enabled
bedrock-provider	not enabled
copilot-provider	not enabled
copilot-acp-provider	not enabled
custom-provider	not enabled
deepseek-provider	not enabled
gemini-provider	not enabled

Plugin	Status
<code>gmi-provider</code>	not enabled
<code>huggingface-provider</code>	not enabled
<code>kilocode-provider</code>	not enabled
<code>kimi-coding-provider</code>	not enabled
<code>minimax-provider</code>	not enabled

Low-Level Surface Files

The repository includes dedicated, low-level public-safe files for each operational surface:

File	Contents
<code>docs/surfaces/tasks.md</code>	Scheduled tasks/cron jobs, modes, scripts, schedules, delivery class, workdir class.
<code>docs/surfaces/skills.md</code>	Skill categories, counts, public-safe examples, operational semantics.
<code>docs/surfaces/hooks.md</code>	Shell hook allowlist state, webhook state, plugin hook manifests.
<code>docs/surfaces/plugins.md</code>	Hermes plugin registry rows and status.
<code>docs/surfaces/mcp-and-toolsets.md</code>	MCP servers and toolset count estimates.
<code>docs/surfaces/models-and-gateway.md</code>	Model routing, gateway status, channel/platform surface.

Model Routing

Role	Provider	Model	Notes
Primary	openai-codex	gpt-5.5	Default for Telegram/API/CLI gateway sessions
Fallback	copilot	gpt-5.4	Used when primary fails
Optional provider	lmstudio	gemma4unc	<code>http://127.0.0.1:1234/v1</code>
Optional provider	nvidia	meta/llama-3.3-70b-instruct	<code>https://integrate.api.nvidia.com/v1</code>
Optional provider	freetimi	cfbt-kimi	<code>http://127.0.0.1:3271/v1</code>
Optional provider	forge_freetimi	cfbt-kimi	<code>http://127.0.0.1:8081/v1</code>
Optional provider	forge_lmstudio	gemma4unc	<code>http://127.0.0.1:8082/v1</code>

Local model trial status

Item	Status
LM Studio endpoint	not reachable at http://127.0.0.1:1234/v1
Reported model IDs	none
Chat smoke test	blocked_or_unavailable: <urlopen error [Errno 111] Connection refused>
Safety decision	Main Hermes remains openai-codex/gpt-5.5 ; local provider is optional until a model can load reliably.

MCP and External Tooling

MCP server	Transport	Public-safe purpose
gbrain	Local HTTP MCP on 127.0.0.1:3131/mcp	Personal knowledge graph, memory, code graph, facts, schema tools.
notebooklm	npx notebooklm-mcp@latest	Grounded research over registered Google NotebookLM notebooks.
codegraph	Local Node command	Code intelligence over selected repositories.

Scheduled Automation

Category	Active jobs	Public-safe purpose
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Current Profiles

The live system currently exposes the default profile publicly as:

Profile	Model	Gateway	Alias	Distribution
◆default	gpt-5.5	running	—	—

Future recommended profile split:

Profile	Purpose	Model stance
default	Main Telegram/personal assistant	openai-codex/gpt-5.5 primary, Copilot fallback.
local-lmstudio	Experimental local-model testing	LM Studio only after a local model can pass smoke tests.
dev	Developer workflows with larger context and coding tools	Can selectively use cloud or local providers.
voice-coach	Low-latency full-duplex voice experiments	Minimal tools/skills, local STT/TTS first.

Reliability and Safety Boundaries

- Public repo does **not** publish secrets, chat IDs, API keys, OAuth tokens, cookies, private finance details, or client-specific payloads.
- The gateway is supervised by systemd and multiple watchdog cron jobs.
- GBrain runs through a shared local HTTP MCP server to avoid PGLite lock contention.
- Backups and sync jobs run through cron with Google Drive/GitHub targets.
- Windows is treated as a remote worker controlled via SSH, not as the primary Hermes control plane.
- Experimental providers are explicitly optional and must pass direct API + Hermes smoke tests before being used for real agent work.

Operational Notes

- Hermes version/status summary:

```
Hermes Agent v0.17.0 (2026.6.19) · upstream 88273002 · local f4a462a2 (+1 carried commit)
Project: ~/.hermes/hermes-agent
Python: 3.11.15
OpenAI SDK: 2.24.0
Update available: 53 commits behind – run 'hermes update'
```

• Fallback chain:

```
Primary:  gpt-5.5  (via openai-codex)

Fallback chain (1 entry):
1. gpt-5.4  (via copilot)

Tried in order when the primary fails (rate-limit, 5xx, connection errors).
Docs: https://hermes-agent.nousresearch.com/docs/user-guide/features/fallback-providers
```

• MCP list:

MCP Servers:

Name	Transport	Tools	Status
codegraph	~/.nvm/versions/no... all		enabled
gbrain	http://127.0.0.1:3131/mcp	all	enabled
notebooklm	npx -y notebooklm-mcp@latest	all	enabled
homeway	https://homeway.io/api/mcp	all	enabled
windows-cua	~/.local/bin/windo... all		enabled
monarch	https://api.monarch.com/mcp	17 selected	enabled
pixelrag	~/github/pixelrag-... all		enabled
vibe_trading	~/.hermes/scripts/... 29 selected		enabled

Maintenance

This repository is regenerated by `scripts/update_architecture.py`. The updater is designed to be safe for a public repository:

- 1. collect local Hermes runtime state,
- 2. redact secrets and private identifiers,
- 3. group sensitive scheduled jobs by category,
- 4. regenerate Markdown, SVG, HTML, JSON inventory, and PDF,

5. commit and push only if files changed.

Source Files

- `ARCHITECTURE.md` - canonical public architecture document.
- `docs/hermes-architecture.svg` - standalone diagram.
- `docs/hermes-architecture.html` - browser-friendly diagram page.
- `docs/Hermes-Architecture.pdf` - rendered PDF copy.
- `docs/surfaces/*.md` - low-level public-safe inventories for tasks, skills, hooks, plugins, MCP/toolsets, and gateway/model routing.
- `data/inventory.public.json` - redacted machine-readable snapshot.
- `scripts/update_architecture.py` - public-safe regeneration script.

Rendered from GBrain markdown via Hermes brain-pdf.